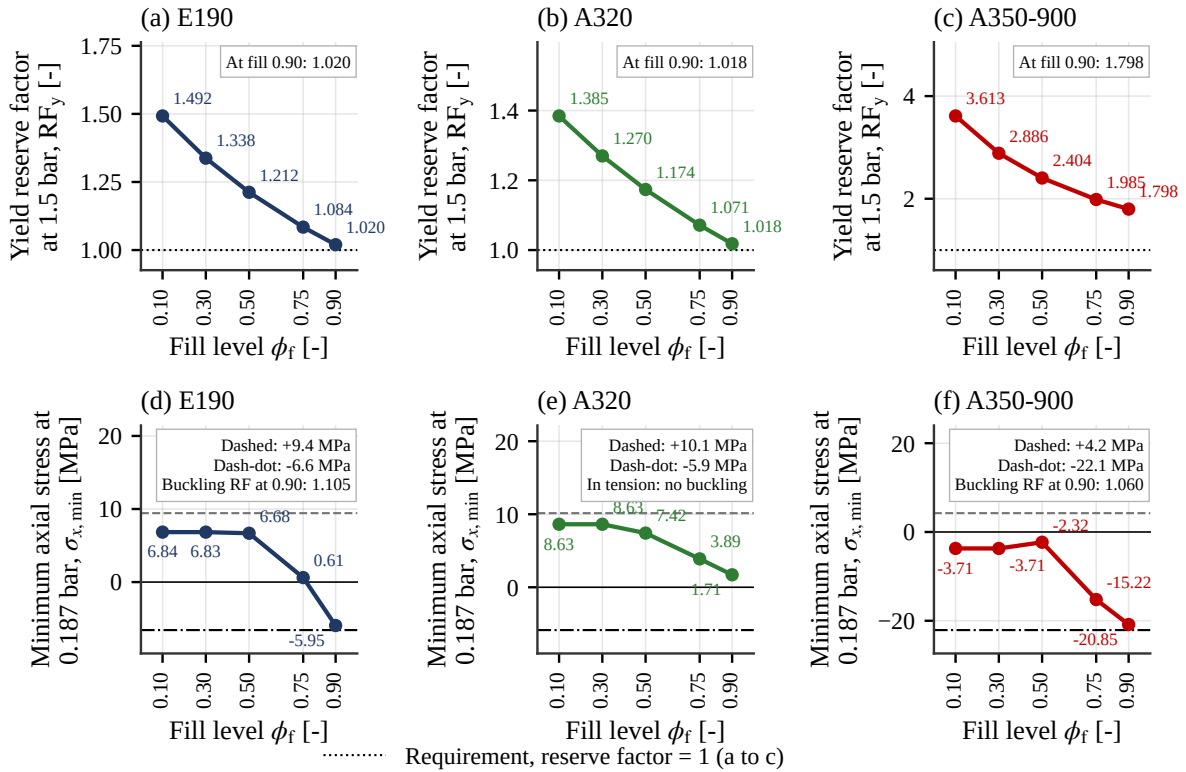
(a) E190
At fill 0.90: 1.020
1.492
1.338
1.212
1.084
1.020
Yield reserve factor at 1.5 bar, RF$_y$ [-]
Fill level ϕ_f [-]
(b) A320
At fill 0.90: 1.018
1.385
1.270
1.174
1.071
1.018
(c) A350-900
At fill 0.90: 1.798
3.613
2.886
2.404
1.985
1.798
(d) E190
Dashed: +9.4 MPa
Dash-dot: -6.6 MPa
Buckling RF at 0.90: 1.105
6.84
6.83
6.68
0.61
-5.95
Minimum axial stress at 0.187 bar, $\sigma_{x,min}$ [MPa]
(e) A320
Dashed: +10.1 MPa
Dash-dot: -5.9 MPa
In tension: no buckling
8.63
8.61
7.42
3.89
1.71
(f) A350-900
Dashed: +4.2 MPa
Dash-dot: -22.1 MPa
Buckling RF at 0.90: 1.060
-3.71
-3.71
-2.32
-15.22
-20.85
Requirement, reserve factor = 1 (a to c)
Axial tension from pressure alone, $pR/2t$ (d to f)
Allowable compression, $-\sigma_{cr}/1.5$ (d to f)